

```

yuktha@LAPTOP-UAVR6HQ0:~$ ls
OSLab
yuktha@LAPTOP-UAVR6HQ0:~$ gedit

** (gedit:539): WARNING **: 08:23:12.974: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:539): WARNING **: 08:23:12.975: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~$ mkdir Practical1
yuktha@LAPTOP-UAVR6HQ0:~$ cd Practical1
yuktha@LAPTOP-UAVR6HQ0:~/Practical1$ gedit practical1.c

** (gedit:1009): WARNING **: 08:25:33.160: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:1009): WARNING **: 08:25:33.160: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~/Practical1$ gcc practical1.c -o practical1
yuktha@LAPTOP-UAVR6HQ0:~/Practical1$ ./practical1
Enter Linux Command: uname

Parent Process
Parent PID : 2294

Child Process
Child PID : 2297
Linux
Child process completed.
yuktha@LAPTOP-UAVR6HQ0:~/Practical1$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            46 bits physical, 48 bits virtual
Byte Order:               Little Endian
CPU(s):                   18
On-line CPU(s) list:      0-17
Vendor ID:                GenuineIntel
Model name:               Intel(R) Core(TM) Ultra 5 125H
CPU family:               6
Model:                    170
Thread(s) per core:       2
Core(s) per socket:       9
Socket(s):                1
Stepping:                 4
BogoMIPS:                 5990.39
Flags:                    fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                          r pge mca cmov pat pse36 clflush mmx fxsr sse s
                          se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                          _tsc rep_good nopl xtopology tsc_reliable nonst
                          op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                          sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                          opcnt tsc_deadline_timer aes xsave avx f16c rdr
                          and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                          brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                          d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                          2 erms invpcid rdseed adx smap clflushopt clwb
                          sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                          vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                          vdiri movdir64b fsrm md_clear serialize ibt flu
                          sh_lld arch_capabilities

```

```

yuktha@LAPTOP-UAVR6HQO: x + v
vdir: movdir64b fsrm md_clear serialize ibt flu
sh_lld arch_capabilities

Virtualization features:
  Virtualization: VT-x
  Hypervisor vendor: Microsoft
  Virtualization type: full

Caches (sum of all):
  L1d: 432 KiB (9 instances)
  L1i: 576 KiB (9 instances)
  L2: 18 MiB (9 instances)
  L3: 18 MiB (1 instance)

NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-17

Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Not affected
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DISABLE
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected

yuktha@LAPTOP-UAVR6HQO:~/Practical1$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro

yuktha@LAPTOP-UAVR6HQO:~/Practical1$ ps
PID TTY TIME CMD
340 pts/0 00:00:00 bash
2331 pts/0 00:00:00 ps

yuktha@LAPTOP-UAVR6HQO:~/Practical1$ top
top - 08:28:41 up 7 min, 1 user, load average: 0.05, 0.04, 0.00
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7743.2 total, 6311.6 free, 637.5 used, 948.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7105.6 avail Mem

```

yuktha@LAPTOP-UAVR6HQ0:~/Practical1\$ top

```
top - 08:28:50 up 8 min, 1 user, load average: 0.05, 0.04, 0.00
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.6 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 7743.2 total, 6313.6 free, 635.6 used, 948.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7107.6 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24160	15368	11580	S	0.0	0.2	0:00.53	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3220	2148	2008	S	0.0	0.0	0:00.10	init
47	root	19	-1	50400	16872	15576	S	0.0	0.2	0:00.20	systemd-journal
79	systemd+	20	0	22416	14508	12012	S	0.0	0.2	0:00.05	systemd-resolve
86	root	20	0	35452	12404	9228	S	0.0	0.2	0:00.41	systemd-udevd
120	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.03	chronyd-starter
125	root	20	0	4472	3132	2884	S	0.0	0.0	0:00.07	cron
131	message+	20	0	8796	5396	4692	S	0.0	0.1	0:00.05	dbus-daemon
164	root	20	0	44096	29692	14556	S	0.0	0.4	0:00.26	networkd-dispat
172	root	20	0	18436	9448	8204	S	0.0	0.1	0:00.13	systemd-logind
174	root	20	0	1792300	15396	12552	S	0.0	0.2	0:00.12	wsl-pro-service
192	syslog	20	0	220548	5948	4656	S	0.0	0.1	0:00.09	rsyslogd
236	_chrony	20	0	23600	11088	7096	S	0.0	0.1	0:00.08	chronyd
248	_chrony	20	0	12096	2468	1824	S	0.0	0.0	0:00.01	chronyd
261	root	20	0	5492	2760	2572	S	0.0	0.0	0:00.01	agetty
287	root	20	0	123460	32712	18104	S	0.0	0.4	0:00.22	unattended-upgr
337	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
338	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.05	Relay(340)
340	yuktha	20	0	6268	5536	3864	S	0.0	0.1	0:00.12	bash
341	root	20	0	8644	5552	4712	S	0.0	0.1	0:00.00	login
390	yuktha	20	0	22764	13468	10988	S	0.0	0.2	0:00.17	systemd
392	yuktha	20	0	22916	4096	2092	S	0.0	0.1	0:00.00	(sd-pam)
422	yuktha	20	0	6136	5468	3916	S	0.0	0.1	0:00.03	bash
543	yuktha	20	0	8704	5268	4724	S	0.0	0.1	0:00.05	dbus-daemon
547	yuktha	20	0	380960	8072	7424	S	0.0	0.1	0:00.01	at-spi-bus-laun
554	yuktha	20	0	8540	5024	4660	S	0.0	0.1	0:00.01	dbus-daemon
555	yuktha	20	0	476876	17836	14948	S	0.0	0.2	0:00.07	xdg-desktop-por
560	yuktha	20	0	304804	6584	6084	S	0.0	0.1	0:00.01	xdg-permission-
565	yuktha	20	0	618996	7560	6872	S	0.0	0.1	0:00.02	xdg-document-po
571	root	20	0	2792	2228	2120	S	0.0	0.0	0:00.00	fusermount3
575	yuktha	20	0	481968	23844	18904	S	0.0	0.3	0:00.09	xdg-desktop-por
583	yuktha	20	0	168688	7860	7204	S	0.0	0.1	0:00.01	at-spi2-registr
954	yuktha	20	0	165220	6480	5996	S	0.0	0.1	0:00.01	dconf-service
2341	yuktha	20	0	8596	6488	3888	R	0.0	0.1	0:00.08	top